

NEELAY RANJAN

Generative Modeling for Safety-Critical, Data-Scarce Domains

@ neelay.ranjan@outlook.com +1 (773) 668-0069 in neelayranjan NeelayRanjan neelayranjan.dev ID 0009-0008-9482-0160

Generative-modeling researcher building diffusion and language-model systems for safety-critical domains where data is scarce and failure is expensive. First-author work on label-efficient vessel segmentation beats SAM with 7 to 22 labels, and work at NASA Ames covers hazard-aware routing and air traffic control speech pipelines.

RESEARCH & WORK EXPERIENCE

NASA Ames Research Center, Aeronautics Directorate

📍 Moffett Field / Mountain View, CA

GenAI Applied Research Intern, SHIFT

📅 Aug 2026 – Present

- Fine-tuned Qwen3-Omni-30B with QLoRA for air traffic control (ATC) speech-to-text on ~1.5 h of real audio (~900 utterances), reaching **17% WER vs. 20% for the existing Whisper ASR** and 30% zero-shot in 5-fold CV.
- Built a generate-and-rank parser for the en route ontology. A text model and rules propose 16 to 32 candidate values per field, and Kev, a typed decision model, picks one. It **raised value-match accuracy from 40% to 80%** over the best rules-based and LLM attempts, at ~90% lower latency and ~400× lower cost than Qwen3 27B.
- Building a text-to-speech synthetic ATC corpus that expands the dataset ~20× (5% real, 95% synthetic). Its outputs feed a maneuver database matched to historical and live FAA airspace records for safety research.

GenAI Applied Research Intern, SLAAC

📅 May – Aug 2026

- Built hazard-aware flight rerouting with diffusion models learned from FAA radar tracks. Guidance applied mid-sampling steers around special-use airspace at inference with no retraining, and **99% of reroutes clear the 25 nm hazard buffer at a median +10 nm (1.1%) added distance**, within 1.2% of the geometric optimum.
- Designed a flight-plan LLM with a custom tokenizer that synthesizes a full day of ~**44,000 FAA-managed flights** matched to historical traffic density, producing simulation-ready traffic days for the software NASA and the FAA use to evaluate future air traffic management (ATM) strategies.

Regenstrief Institute

📍 Indianapolis, IN

Applied Vision-AI Researcher

📅 Feb 2024 – Present

- Showed an x0 flipped-objective diffusion segmenter outperforms SAM (ViT-B) on benchmark angiograms in ultra-low-data regimes, reaching **88.2% Dice with just 16 labels**. Its pre-labels bootstrap surgeon labeling and **cut correction time by 75%**.
- Built a modular SAM-x0 tiled refinement pipeline (SAM prior + partial-resolution x0 diffusion refiner) for real-world, region-agnostic angiograms at native 1024 resolution. It reaches **70% Dice with 22 labeled images**, beating both SAM (46%) and raw x0 (32%) at a fraction of full-resolution compute.
- Extended the pipeline to cerebral moyamoya MRA, segmenting vasculature from **only 7 labeled images** with patient-level cross-validation and tiling augmentation, at ~55% Dice vs. SAM's ~20%.

SELECTED PUBLICATIONS

- N. Ranjan, S. Clarke, et al. “Generate-and-rank extraction of air traffic control clearances into an en route ontology.” *NASA Ames. Extended abstract for AIAA AVIATION 2027, in preparation.*
- N. Ranjan et al. “Bootstrapping surgeon labeling campaigns with x0-diffusion: label-efficient vessel segmentation of catheter-based angiograms.” *First author, in preparation.*
- Co-author, “Helical Antenna for Electromagnetic Field Stimulation in Alzheimer’s Disease Therapy.” *Presented at IEEE MWSCAS 2026. Led the PCB design team.*

PROJECTS & APPLIED RESEARCH

Solar-Powered Energy-Based Chess Engine

📅 2025

- Trained an energy-based model that scores and ranks moves from a learned energy landscape over game states, reaching ~**1900–2200 estimated Elo** against a ladder of rated Stockfish depths.
- Embedded the EBM on a Raspberry Pi Zero 2 W (~1 s per move) as a standalone chess engine **powered solely by onboard solar & supercapacitors**. Quantization tests showed FP32 beat INT8 on ARM, which had no native INT8 kernels.

LLM Design Team Lead, V2X Aircraft Maintenance Assistant

📅 Aug 2024 – May 2025

- Led design and fine-tuning of a maintenance-assistant LLM for V2X, a Raytheon divestiture. Its two-stage RAG pipeline picks sources from their summaries, then pinpoints and quotes the relevant section, **cutting hallucinations from ~40% to ~5%**.

KEY RESULTS



17% WER vs. 20%

QLoRA ATC speech-to-text, 5-fold CV

80% value match vs. 40%

Generate-and-rank parsing with Kev

99% hazard clearance

Diffusion reroutes at +1.1% distance



88.2% Dice at 16 labels

x0 segmenter vs. SAM on angiograms

75% faster correction

x0 pre-labels speed surgeon labeling

70% Dice at 1024 px

Real-world angiograms (SAM 46%)

7 training images

Moyamoya MRA at ~55% (SAM ~20%)



5% hallucinations vs. 40%

Two-stage RAG, maintenance LLM

EDUCATION

Purdue University, College of Engineering

B.S. in Artificial Intelligence

📅 Expected May 2027 📍 Indianapolis, IN

Concentration: Intelligent Control & Systems

Minor: Mathematics

GPA: 3.68 / 4.0 · Dean’s List

John Martinson Honors College

CITI: Biomedical Research, RCR, GCP

TECHNICAL SKILLS

Generative modeling

foundation-model adaptation (SAM)
diffusion (x0 / v-prediction) transformers
energy-based models LLM fine-tuning & RAG
safety-critical AI interpretability
multimodal pipelines

ML engineering

Python C/C++ CUDA PyTorch Git
medical image segmentation Docker W&B
Linux / cluster administration quantization
transfer learning Slurm / H100

Hardware / edge

PCB design ultra-low-power inference
embedded ML